

Introduction to instantons in gauge theories and gravity

Cristóbal Corral^{1,*}

¹*Departamento de Ciencias, Facultad de Artes Liberales, Universidad Adolfo Ibáñez,
Avenida Padre Hurtado 750, 2562340, Viña del Mar, Chile*

Abstract

These notes were presented at the *GravUC School 2024: Topics in Quantum Gravity*, held in Pontificia Universidad Católica de Chile, Santiago, December 16th to 20th, 2025. They aim to introduce topologically nontrivial configurations in gauge theories and gravity, such as monopoles, dyons, merons, and instantons, in a modern language of theoretical physics. Additionally, the contribution to nonperturbative effects, such as anomalies in quantum field theories, is presented following Fujikawa's method. Finally, we present examples of gravitational instantons in anti-de Sitter space and discuss their similarities with their Yang-Mills counterpart.

* crislobal.corral@uai.cl

I. INTRODUCTION

The connection between topology and quantum effects is perhaps one of the most fascinating discoveries in physics. It bridges different areas of mathematical physics, topology, high-energy theory, and condensed matter physics, among others, generating interesting interdisciplinary scientific collaborations. For instance, superconductivity, topological insulators, tunneling, quantum impurities, etc., are nonperturbative effects in essence, and they are usually described by states of matter whose nontrivial topology protects them from decaying into the trivial vacuum of the theory. Topological solitons, for instance, are well-known examples of the latter, representing pseudoparticle states with finite energy that saturate a Bogomol'nyi–Prasad–Sommerfield bound. This property allows one to reduce the second-order field equations down to a first-order differential equation whose solution automatically satisfies the field equations. Among these configurations, one can mention magnetic monopoles, Skyrmons, vortices, and Yang-Mills instantons, whose nontrivial topology in nonlinear theories produces interesting effects that their linear counterparts do not exhibit.

Since gravity is a nonlinear theory that can be considered as a gauge theory for the $SO(4)$ group in Euclidean signature, it is natural to ask whether it admits the gravitational analog of topological solitons. Several of these solutions has been studied in the last years, e.g. gravitational solitons, vortices, and instantons, bearing close resemblance with those in Yang-Mills theory. Gravitational instantons, for instance, are (anti-)self-dual stationary regular solutions to Euclidean gravity with nontrivial topology. An interesting example of this class can be obtained from the analytic continuation of the Taub-Newman-Tamburino-Unti metric [1, 2]. The original geometry was obtained as the $U(1)$ -fibration on 2-dimensional Kähler manifolds [3] and it was further generalized to arbitrary D dimensions in Ref. [4]. Depending on whether its set of fixed points is zero or $(D - 2)$ -dimensional [5], the solution is said to have a nut or a bolt, respectively [3]. In four dimensions, the previous conditions imply that the spacetime becomes globally (anti-)self-dual or asymptotically (anti-)self-dual, respectively. This solution is endowed with a new parameter—the NUT charge—related to the first Chern number at infinity [6]. The latter sources the magnetic part of the Weyl tensor [7, 8], and it is closely related to the existence of the Misner string [9]. In AdS, the (anti-)self-dual configuration saturates the BPS bound and the renormalized Euclidean on-shell action is proportional to the Chern-Pontryagin index [7, 8, 10],

capturing important features of instantons in Yang-Mills theory.

Additionally, these topologically nontrivial configurations can break classical symmetries at the quantum level. The axial anomaly, for instance, is a concrete example of the latter. It occurs when the axial symmetry of a classical system is broken due to quantum effects. This phenomenon was first observed in the anomalous decay of pions which would have been otherwise classically forbidden according to the standard model of particle physics [11–16]. Its origin can be understood in terms of the interaction between spinors and background gauge fields with nontrivial topology as it usually happens with instantons [17, 18]. Indeed, the existence of the latter implies that the physical vacuum must be a linear combination of topological states labeled by their winding numbers [19]. This effect can be encoded in the Yang-Mills action by adding the Pontryagin density, i.e. the θ -term [20, 21], whose physical consequences have been studied in detail in Refs. [22–27]. Another way to determine the axial anomaly is the Atiyah-Patodi-Singer theorem [28, 29] which provides a deep connection between the difference of chiral fermions and topological invariants of the Pontryagin class. Equivalently, this phenomenon can be thought of as a nonperturbative effect that arises from the nontrivial transformation of the fermionic measure under a $U(1)$ -axial transformation in the path-integral formulation of gauge theories [30, 31].

These notes are intended to introduce instantons in gauge theories and gravity to graduate students in the early stages of their careers, providing them with modern tools in theoretical high-energy physics for doing research. The content is organized as follows: in the first part, topological solitons, e.g. monopoles, dyons, instantons, and merons, in gauge theories are reviewed. In the second part, we study global symmetries of the massless Dirac action and show how the axial anomaly appears following Fujikawa’s approach. In the last part, we present gravitational instantons and (anti-)self-duality in AdS and show how these configurations bear a close resemblance with instantons in Yang-Mills theory.

II. GAUGE THEORIES

Gauge theories are the best description of interactions that we have so far. The standard model of particle physics, for instance, can be formulated as a gauge theory for the $SU(3) \times SU(2) \times U(1)$ group. The latter is broken down to $SU(2) \times U(1)$ group after the spontaneous symmetry breaking

is triggered by the nontrivial vacuum expectation value of the Higgs field. General relativity, on the other hand, can be thought of as a gauge theory for the Lorentz group, which, in the Euclidean section, is represented by the $SO(4)$ group. However, besides its local symmetries, gravity also has a diffeomorphism invariance, which renders its gauge theory interpretation more challenging. Still, understanding interactions requires learning gauge theories in detail. Here, we introduce its elementary aspects,

Let $U(x)$ be an element of a Lie group G with Lie algebra $\mathfrak{g}$. The latter is spanned by the generators T_I , satisfying the Lie algebra $[T_I, T_J] = f_{IJK}T^K$, where f_{IJK} are the structure constants and internal indices are lowered and raised with the Cartan-Killing metric $g_{IJ} = \text{tr}(T_I T_J)$ and its inverse g^{IJ} , respectively. Now, consider a field, e.g. $\Psi(x)$, in the fundamental representation of G . Then, one can show that the ordinary partial derivative, $\partial_\mu \Psi$ does not transform covariantly under a local transformation $\Psi(x) \rightarrow \Psi'(x) = U(x)\Psi(x)$, since

$$\partial_\mu \Psi \rightarrow \partial_\mu \Psi' = \partial_\mu [U(x)\Psi(x)] = \partial_\mu U(x)\Psi(x) + U(x)\partial_\mu \Psi(x), \quad (1)$$

where the first term in the last equality spoils covariance. To circumvent this problem, one promotes the ordinary partial derivative to be a new one, say

$$\partial_\mu \rightarrow D_\mu = \partial_\mu + A_\mu(x), \quad (2)$$

where $A_\mu(x) = A_\mu^I(x)T_I$ is dubbed gauge connection. The latter belongs to the adjoint representation of G and it transforms as a 1-form under diffeomorphisms. For the new derivative $D_\mu \Psi(x)$ to transform covariantly under the action of G , it should satisfy

$$D_\mu \Psi(x) \rightarrow (D_\mu \Psi(x))' = U(x)D_\mu \Psi(x) = U(x)(\partial_\mu + A_\mu)\Psi. \quad (3)$$

So, we need to demand a suitable transformation of $A_\mu(x) \rightarrow A'_\mu(x)$ such that it cancels the inhomogeneous part of Eq. (1). This can be achieved by comparing¹

$$D_\mu \Psi \rightarrow (D_\mu \Psi)' = (\partial_\mu + A'_\mu)\Psi' = (\partial_\mu + A'_\mu)U\Psi = \partial_\mu U\Psi + U\partial_\mu \Psi + A'_\mu U\Psi, \quad (4)$$

with the right-hand side of Eq. (3); this yields

$$U\partial_\mu \Psi + UA_\mu \Psi = \partial_\mu U\Psi + U\partial_\mu \Psi + A'_\mu U\Psi \quad (5)$$

$$(UA_\mu - \partial_\mu U - A'_\mu U)\Psi = 0. \quad (6)$$

¹ From hereon, we omit the coordinate dependence of $U(x)$ and $\Psi(x)$.

Then, for an arbitrary Ψ and multiplying from the right with U^{-1} , such that $U^{-1}U = UU^{-1} = \mathbb{I}$, we obtain that, in order for $D_\mu \Psi$ to transform covariantly under the action of G , the gauge connection should transform as $A_\mu \rightarrow A'_\mu$ where

$$A'_\mu = UA_\mu U^{-1} - \partial_\mu U U^{-1} = UA_\mu U^{-1} + U\partial_\mu U^{-1} = U(\partial_\mu + A_\mu)U^{-1} = UD_\mu U^{-1}, \quad (7)$$

where in the last equality we have used $\partial_\mu(UU^{-1}) = 0$ and the Leibniz rule. If the gauge potential is $A_\mu = U\partial_\mu U^{-1}$, we say this configuration is *pure gauge*. Indeed, one can see that the non-Abelian field strength — see Eq. (8) below — vanishes identically when evaluated at this solution.

The non-Abelian field strength (or gauge curvature) defines the dynamics of the gauge connection. It is defined as the commutator of covariant derivatives acting on a nontrivial representation of G , that is,

$$\begin{aligned} [D_\mu, D_\nu]\Psi &= D_\mu D_\nu \Psi - D_\nu D_\mu \Psi = (\partial_\mu + A_\mu)(\partial_\nu + A_\nu)\Psi - (\partial_\nu + A_\nu)(\partial_\mu + A_\mu)\Psi \\ &= \partial_\mu A_\nu \Psi + A_\nu \partial_\mu \Psi + A_\mu \partial_\nu \Psi + A_\mu A_\nu \Psi - (\mu \leftrightarrow \nu) \\ &:= F_{\mu\nu} \Psi \end{aligned}$$

where the gauge field strength is defined as

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]. \quad (8)$$

The commutator reveals the non-Abelian nature of the gauge connection and is responsible for its self-interaction; at the classical level, this is forbidden in Abelian gauge theories like electromagnetism. A direct consequence of this construction is that the gauge field strength satisfies the Bianchi identity, that is,

$$D_{[\lambda} F_{\mu\nu]} = 0, \quad \text{where} \quad D_\mu = \partial_\mu + [A_\mu, \cdot] \quad (9)$$

is the gauge covariant derivative in the adjoint representation.

In four dimensions, one can use the Levi-Civita tensor $\varepsilon_{\mu\nu\lambda\rho}$ to define the dual non-Abelian field strength, that is,

$$\tilde{F}_{\mu\nu} := \frac{1}{2} \varepsilon_{\mu\nu\lambda\rho} F^{\lambda\rho}. \quad (10)$$

Then, the Bianchi identity can be equivalently written as $D_\mu \tilde{F}^{\mu\nu} = 0$. Indeed, there exists a special class of configurations which are called *(anti-)self-dual*, satisfying the first-order relation

$$F_{\mu\nu} = \pm \tilde{F}_{\mu\nu} . \quad (11)$$

As we will see, these configurations possess important properties, e.g. they saturate a Bogomol'nyi–Prasad–Sommerfield bound and their energy-momentum tensor vanishes identically on Riemannian manifolds. Even more, vacuum Maxwell equations are invariant under (anti-)self-duality transformations $F_{\mu\nu} \rightarrow \pm \tilde{F}_{\mu\nu}$; this is a broken symmetry in the presence of sources.

Having reviewed all the basic ingredients and fixed notation for general non-Abelian gauge theories, we can now discuss their dynamics and topological properties. We will start from the simplest Abelian case, whose best-known example is Maxwell's electromagnetic theory.

Exercises II

1. Show that the field strength vanishes for a pure gauge potential.
2. Using the generators' Lie algebra, show that the gauge field strength (8) can be spanned in its basis as $F_{\mu\nu} = F_{\mu\nu}^I T_I$, where

$$F_{\mu\nu}^I = \partial_\mu A_\nu^I - \partial_\nu A_\mu^I + f_{JK}^I A_\mu^J A_\nu^K . \quad (12)$$

3. Show that the field strength can be written in differential forms as

$$F = \frac{1}{2} F_{\mu\nu}^I T_I dx^\mu \wedge dx^\nu = dA + \frac{1}{2} [A, A] = dA + A \wedge A , \quad (13)$$

where $A = A_\mu^I T_I dx^\mu$ is the 1-form gauge potential and d is the exterior derivative.

4. Show that, under a gauge transformation $A \rightarrow A' = UAU^{-1} + U dU^{-1}$, the 2-form field strength transforms covariantly, that is, $F \rightarrow F' = U F U^{-1}$.
5. Using differential forms, show that the Bianchi identity is $DF = 0$, where $D = d + [A, \cdot]$ is the exterior gauge-covariant derivative in the adjoint representation.

III. MAXWELL THEORY

Maxwell's electrodynamics can be formulated as a gauge theory for the $U(1)$ group. In this case, the gauge transformation of a complex field Ψ in the fundamental representation of $U(1)$ can be conveniently expressed as $\Psi \rightarrow \Psi' = e^{i\alpha(x)}\Psi$. Then, the covariant derivative $D_\mu = \partial_\mu + iA_\mu$ transforms as

$$D_\mu \Psi \rightarrow (D_\mu \Psi)' = e^{i\alpha(x)} D_\mu \Psi, \quad \text{if} \quad A_\mu(x) \rightarrow A'_\mu(x) = A_\mu(x) + \nabla_\mu \alpha(x), \quad (14)$$

where $\alpha : \mathcal{M} \rightarrow \mathbb{R}$ is a real scalar function. Since the gauge group is Abelian, the commutator in Eq. (8) vanishes, and the $U(1)$ field strength becomes $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

In the absence of external sources, the Maxwell Lagrangian can be written on a four-dimensional Riemannian manifold $(\mathcal{M}, g)$ as

$$I_M[A_\mu] = \frac{1}{4} \int_{\mathcal{M}} d^4x \sqrt{|g|} F_{\mu\nu} F^{\mu\nu} := \int_{\mathcal{M}} d^4x \sqrt{|g|} \mathcal{L}_M, \quad (15)$$

where $g = \det g_{\mu\nu}$ is the metric determinant. The Maxwell equations can be obtained by performing stationary variations of the action (15) with respect to the gauge potential A_μ , giving

$$\nabla_\mu F^{\mu\nu} = 0, \quad (16)$$

where ∇ is the connection compatible with $g_{\mu\nu}$. Additionally, in theories where the metric is dynamical, e.g. general relativity, the stress-energy tensor of Maxwell fields is defined as the variation of the matter action with respect to the metric, giving

$$T_{\mu\nu} := -\frac{2}{\sqrt{|g|}} \frac{\delta(\sqrt{|g|} \mathcal{L}_M)}{\delta g^{\mu\nu}} = F_{\mu\lambda} F_\nu{}^\lambda - \frac{1}{4} g_{\mu\nu} F_{\lambda\rho} F^{\lambda\rho}. \quad (17)$$

Diffeomorphism invariance of the matter action implies that the latter is conserved on-shell, i.e. $\nabla_\mu T^{\mu\nu} = 0$. Notice that the trace of the stress-energy tensor of Maxwell fields vanishes identically in four dimensions. This is a consequence of the invariance of the action under simultaneous Weyl rescalings $g_{\mu\nu} \rightarrow g'_{\mu\nu} = e^{2\Omega(x)} g_{\mu\nu}$ and $A_\mu \rightarrow A'_\mu = A_\mu$.

In (pseudo-)Riemannian geometry, one can foliate the space using codimension-1 hypersurfaces with unit-normal vector $n = n^\mu \partial_\mu$, where $n \cdot n = \sigma = \pm 1$. This foliation defines the induced metric $h_{\mu\nu} = g_{\mu\nu} - \sigma n_\mu n_\nu$. Then, one can decompose the Faraday tensor in terms of its electric and

magnetic parts, that is,

$$E^\mu = F^{\mu\nu} n_\nu \quad \text{and} \quad B^\mu = \tilde{F}^{\mu\nu} n_\nu, \quad (18)$$

respectively, whose definition implies $n_\mu E^\mu = n_\mu B^\mu = 0$.

A. Conserved charges

The conserved charges associated with Maxwell fields can be defined using a Gauss law. Consider a hypersurface Σ with induced metric $h_{\mu\nu}$ and the definition of unit-normal given above. Now, in the presence of a conserved current, i.e. $\nabla_\mu J^\mu = 0$, integration over $\mathcal{M}$ with boundary $\partial\mathcal{M}$ yields

$$0 = \int_{\mathcal{M}} d^4x \sqrt{|g|} \nabla_\mu J^\mu = \oint_{\partial\mathcal{M}} d^3x \sqrt{|h|} n_\mu J^\mu = \oint_{\partial\mathcal{M}} J^\mu d\Sigma_\mu \quad (19)$$

where h denotes the determinant of the induced metric and $d\Sigma_\mu$ is the oriented volume element of the codimension-1 boundary $\partial\mathcal{M}$. This implies that, if the closed boundary can be written in terms of two hypersurfaces, say $\partial\mathcal{M} = \partial\mathcal{M}_1 \cup \partial\mathcal{M}_2$, the integrals on each hypersurface are equal, modulo orientation, since

$$0 = \oint_{\partial\mathcal{M}} J^\mu d\Sigma_\mu = \int_{\partial\mathcal{M}_1} J^\mu d\Sigma_\mu^{(1)} + \int_{\partial\mathcal{M}_2} J^\mu d\Sigma_\mu^{(2)} \rightarrow \int_{\partial\mathcal{M}_1} J^\mu d\Sigma_\mu^{(1)} = - \int_{\partial\mathcal{M}_2} J^\mu d\Sigma_\mu^{(2)} \quad (20)$$

Then, one can define a conserved charge that is independent of the codimension-1 hypersurface we are considering, that is,

$$Q := \int_{\Sigma} J^\mu d\Sigma_\mu = \int_{\Sigma} d^3x \sqrt{|h|} n_\mu J^\mu. \quad (21)$$

Taking the covariant derivative on the Maxwell equation (16), i.e. $\nabla_\mu \nabla_\nu F^{\mu\nu} = 0$, one can see that it allows one to define a divergenceless vector, i.e. $J^\mu = \nabla_\nu F^{\mu\nu}$, whose associated conserved charge is constant on any Σ . Then, using the identity $n_\mu \nabla_\nu \omega^{\mu\nu} = \mathcal{D}_\nu (n_\mu \omega^{\mu\nu})$, where $\omega^{\mu\nu} = -\omega^{\nu\mu}$ is an arbitrary two-form and $\mathcal{D}_\mu$ the covariant derivative compatible with $h_{\mu\nu}$, we find that the integral in Eq. (21) can be written as

$$\begin{aligned} Q &= \int_{\Sigma} d^3x \sqrt{|h|} n_\mu \nabla_\nu F^{\mu\nu} = \int_{\Sigma} d^3x \sqrt{|h|} \mathcal{D}_\mu (n_\nu F^{\mu\nu}) \\ &= \oint_{\partial\Sigma} d^2x \sqrt{|\sigma|} n_\mu u_\nu F^{\mu\nu} = \oint_{\partial\Sigma} F^{\mu\nu} d\Sigma_{\mu\nu}, \end{aligned} \quad (22)$$

where $\sigma_{\mu\nu} = h_{\mu\nu} - u_\mu u_\nu$ is the induced metric on the codimension-2 boundary $\partial\Sigma$, whose oriented volume element is denoted by $d\Sigma_{\mu\nu}$. Upon a suitable normalization, this conserved quantity represents the electric charge.

It is worth mentioning that a similar construction can be achieved from the Maxwell action's $U(1)$ gauge invariance. This can be seen by evaluating the variation in Eq. (30) on a solution of the Maxwell equations (16). Then, invariance of I_{MCS} under $U(1)$ gauge transformations of the form $\delta A_\mu = \nabla_\mu \alpha$ yields $\nabla_\mu J^\mu = 0$, where J^μ can be written explicitly from the last term of Eq. (30). Thus, upon normalization of $\alpha(x)$ at infinity, a conserved charge associated with the $U(1)$ invariance leads to the same conserved charge as in Eq. (22). Finally, as the Bianchi identity can be written as $\nabla_\mu \tilde{F}^{\mu\nu} = 0$, a similar definition can be made for the magnetic charge, that is,

$$\tilde{Q} = \oint_{\partial\Sigma} \tilde{F}^{\mu\nu} d\Sigma_{\mu\nu} \quad (23)$$

In this case, the codimension-2 integral becomes the index of the first Chern class, providing a natural Dirac quantization condition from topological considerations; we will return to this point when studying the magnetic monopole.

B. (Anti-)self-duality

As mentioned in Sec. II, there exists a special class of field configurations in Riemannian manifolds known as (anti-)self-dual gauge fields; they satisfy $F_{\mu\nu} = \pm \tilde{F}_{\mu\nu}$. Indeed, they solve the Maxwell equations (16) by the Bianchi identity $\nabla_{[\mu} F_{\nu\lambda]} = 0$, since

$$\nabla_\mu F^{\mu\nu} = \pm \nabla_\mu \tilde{F}^{\mu\nu} = \pm \frac{1}{2} \varepsilon^{\mu\nu\lambda\rho} \nabla_\mu F_{\lambda\rho} = 0. \quad (24)$$

Thus, the second-order linear Maxwell equations can be reduced to a first-order one by imposing (anti-)self-duality. Moreover, these configurations minimize the action (15). This can be seen by noticing that Eq. (15) can be rewritten as

$$I_{\text{M}} = \frac{1}{8} \int_{\mathcal{M}} d^4x \sqrt{|g|} \left(F_{\mu\nu} \pm \tilde{F}_{\mu\nu} \right) \left(F^{\mu\nu} \pm \tilde{F}^{\mu\nu} \right) \mp \frac{1}{4} \int_{\mathcal{M}} d^4x \sqrt{|g|} \tilde{F}_{\mu\nu} F^{\mu\nu}. \quad (25)$$

The last piece is proportional to a gauge-invariant topological term of the Pontryagin class. It is related to the difference between dual and anti-self dual harmonic forms in the corresponding

dimension. For Abelian $U(1)$ fields, it is defined as

$$C_2[A] = \frac{1}{16\pi^2} \int_{\mathcal{M}} d^4x \sqrt{|g|} \tilde{F}_{\mu\nu} F^{\mu\nu}, \quad (26)$$

and it can be written as a boundary term since $\tilde{F}_{\mu\nu} F^{\mu\nu} = \nabla_\mu K^\mu$, where $K^\mu = \varepsilon^{\mu\nu\lambda\rho} A_\nu F_{\lambda\rho}$ is known as the Chern-Simons form for the $U(1)$ group and $\varepsilon^{\mu\nu\lambda\rho}$ is the Levi-Civita tensor. Although the Pontryagin density is invariant under $U(1)$ gauge transformations [cf. Eq. (14)], the Chern-Simons form is not, since it transforms as $K^\mu \rightarrow K'^\mu = K^\mu + \nabla_\nu \alpha^{\mu\nu}$, where $\alpha^{\mu\nu} = \alpha(x) \varepsilon^{\mu\nu\lambda\rho} F_{\lambda\rho}$.

Using the fact that the first term of the right-hand side of Eq. (25) must be positive definite in Euclidean signature, we conclude that Eq. (25) can be written as

$$I_M \pm 4\pi^2 C_2[A] = \frac{1}{8} \int_{\mathcal{M}} d^4x \sqrt{|g|} \left(F_{\mu\nu} \pm \tilde{F}_{\mu\nu} \right) \left(F^{\mu\nu} \pm \tilde{F}^{\mu\nu} \right) \geq 0. \quad (27)$$

Thus, it becomes clear that the bound is saturated for (anti-)self-dual configurations, where the action becomes proportional to the Chern-Pontryagin index of $U(1)$, that is, $I_M = \mp 4\pi^2 C_2[A]$. This is known as the Bogomol'nyi-Prasad-Sommerfield (BPS) bound.

Indeed, the Maxwell action (15) can be supplemented by the Chern-Pontryagin topological term from scratch without modifying the Maxwell equations (16), namely,

$$I_{\text{MCS}}[A_\mu] = \frac{1}{4} \int_{\mathcal{M}} d^4x \sqrt{|g|} F_{\mu\nu} F^{\mu\nu} + \frac{\theta}{4} \int_{\mathcal{M}} d^4x \sqrt{|g|} \tilde{F}_{\mu\nu} F^{\mu\nu}. \quad (28)$$

Here, θ is a dimensionless constant known as the θ -parameter. This term, however, does modify the finite part of the action and variations thereof, changing the value of conserved charges. Indeed, if $\theta = \pm 1$, this term cancels the last term in Eq. (25) and the action in Eq. (28) can be written as

$$I_{\text{MCS}}[A_\mu] = \frac{1}{8} \int_{\mathcal{M}} d^4x \sqrt{|g|} \left(F_{\mu\nu} \pm \tilde{F}_{\mu\nu} \right) \left(F^{\mu\nu} \pm \tilde{F}^{\mu\nu} \right). \quad (29)$$

Then, it is clear that adding the Pontryagin term with a fixed θ -parameter changes the ground state of the theory from pure-gauge configurations to the whole family of (anti-)self-dual solutions. Indeed, (anti-)self-dual configurations also represent stationary points of the Euclidean action. This can be seen by noticing that variation of the action with respect to the $U(1)$ gauge potential vanishes for this class of solutions, since

$$\delta I_{\text{MCS}} = - \int_{\mathcal{M}} d^4x \sqrt{|g|} \delta A_\nu \nabla_\mu F^{\mu\nu} + \int_{\mathcal{M}} d^4x \sqrt{|g|} \nabla_\mu \left[\delta A_\nu (F^{\mu\nu} + \theta \tilde{F}^{\mu\nu}) \right], \quad (30)$$

which is identically zero for (anti-)self-dual configurations if and only if $\theta = \pm 1$.

As a final remark of this Subsection, it is worth mentioning that, in Riemannian spaces, the stress-energy tensor of Maxwell fields in Eq. (17) can be rewritten as

$$T_{\mu\nu} = \frac{1}{2} g^{\lambda\rho} \left(F_{\mu\lambda} + \tilde{F}_{\mu\lambda} \right) \left(F_{\nu\rho} - \tilde{F}_{\nu\rho} \right). \quad (31)$$

Therefore, it is clear that the latter vanishes identically if $F_{\mu\nu} = \pm \tilde{F}_{\mu\nu}$. When coupled to gravity, this implies that nontrivial (anti-)self-dual configurations do not backreact; they resemble stealth configurations in scalar-tensor theories [32, 33].

C. The magnetic monopole

Although the magnetic monopole has not been observed in nature so far, its mere existence would imply the celebrated Dirac quantization condition of the magnetic charge [34]. This configuration is a simple yet topologically nontrivial solution to Maxwell's equations, whose vector potential that generates the corresponding magnetic field for a point-like source located at the origin is

$$A = A_\mu dx^\mu = -p \cos \vartheta d\varphi \quad \rightarrow \quad F = dA = p \sin \vartheta d\vartheta \wedge d\varphi, \quad (32)$$

where p is the magnetic charge. However, introducing a magnetic monopole as a point charge would modify the Maxwell equation $\vec{\nabla} \cdot \vec{B} \neq 0$, which, in turn, spoils the Bianchi identity. Even more, in Euclidean four-dimensional space, the gauge potential in Eq. (32) is singular, since

$$F_{\mu\nu} F^{\mu\nu} = \frac{2p^2}{r^4} \quad \text{and} \quad \check{A} = A^\mu \partial_\mu = -\frac{p \cos \vartheta}{r^2 \sin^2 \vartheta} \partial_\varphi \quad (33)$$

where $A^\mu = g^{\mu\nu} A_\nu$. Thus, one concludes that the gauge potential for the Dirac monopole is singular at $r = 0$, and it is endowed with a string-like singularity at $\vartheta = 0, \pi$; this is called the Dirac string.

To circumvent this problem, Wu-Yang worked out the modern approach to the magnetic monopole, using the nontrivial topology of fiber bundles induced by this configuration [35, 36]. Instead of considering a point-like magnetic source located at the origin of $\mathbb{R}^3$ which produces a string-like singularity on the gauge potential and spoils the Bianchi identity as Dirac did, Wu-Yang considered removing a point where the magnetic monopole was located, such that the topology of

the space is $\mathbb{R}^3/\{0\} = \mathbb{S}^2$. Then, they considered two Abelian gauge potentials one-forms $A_{\pm}$, one on each hemisphere $\mathbb{S}_{\pm}^2$, such that $\mathbb{S}^2 = \mathbb{S}_{+}^2 \cup \mathbb{S}_{-}^2$. These two patches overlap on the equator, representing the two hemispheres' boundary. Then, one can define a regular gauge potential 1-form on $\mathbb{S}_{\pm}^2$ that leads to exactly the same Maxwell field strength as the magnetic monopole in Eq. (32), that is,

$$A_{\pm} = -p(\cos \vartheta \mp 1) d\varphi \quad \rightarrow \quad F = dA_{\pm} = p \sin \vartheta d\vartheta \wedge d\varphi, \quad (34)$$

This choice guarantees that the dual vector fields associated with these potential one-forms are regular at $\vartheta = 0$ and $\vartheta = \pi$, since

$$\tilde{A}_{\pm} = A_{\pm}^{\mu} \partial_{\mu} = -\frac{p(\cos \vartheta \mp 1)}{r^2 \sin^2 \vartheta} \partial_{\varphi} \quad \rightarrow \quad A_{\pm}^{\varphi} \approx \pm \frac{p}{2r^2} \quad \text{as} \quad \vartheta = 0, \pi. \quad (35)$$

Notice that $A_{\pm}$ yields the same electromagnetic field. Therefore, they must be related by a gauge transformation. Indeed, at the equator, these two $U(1)$ potentials are related by

$$A_{+} = A_{-} + 2p d\varphi. \quad (36)$$

Recall that, in the previous subsection, we show that the magnetic charge can be computed from the codimension-2 integral of the field strength two-form. In this case, however, we have separated the integration region into two hemispheres, each of them with a gauge potential one-form $A_{\pm}$. Then, the magnetic charge can be written as

$$\begin{aligned} \tilde{Q} &= \oint_{\mathbb{S}^2} F = \int_{\mathbb{S}_{+}^2} dA_{+} + \int_{\mathbb{S}_{-}^2} dA_{-} = \oint_{\partial\mathbb{S}_{+}^2} A_{+} + \oint_{\partial\mathbb{S}_{-}^2} A_{-} = \oint_{\gamma} (A_{+} - A_{-}) \\ &= 2p \int_0^{2\pi} d\varphi = 4\pi p. \end{aligned} \quad (37)$$

Using the fact that the first Pontryagin index can be written as $C_1[\Sigma] = -\frac{1}{2\pi} \int_{\Sigma} F$, we conclude that standard topological arguments must quantize the magnetic charge. This is a simple yet nontrivial example of topologically nontrivial solutions in gauge theories.

D. (Anti-)Self-dual dyons

Despite the interesting topological properties and the quantization condition on the magnetic monopole's charge, its field strength does not remain invariant under self-duality transformations. However, a particular class of solutions known as dyons can be rendered (anti-)self-dual upon suitable conditions on the electric and magnetic charges. In this section, we study the main aspects of these configurations.

Dyons are field configurations having both electric and magnetic fields. They were first proposed by Schwinger as a simple model for quarks [37], allowing him to predict the existence of the J/ψ meson, whose discovery in 1974 led Burton Richter and Samuel Ting to win the Nobel Prize in Physics in 1976. Indeed, electric (magnetic) monopoles can be obtained as the zero magnetic (electric) charge limit of dyons, and in theories possessing CP invariance, these two charges are quantized according to the Dirac–Schwinger–Zwanziger relation. Dyons are predicted by grand-unified theories, and, similar to the magnetic monopole, their gauge potential does not respect the symmetries of the background, although their field strength does.

The simplest example of dyons can be found in Maxwell's theory on four-dimensional Euclidean space. For instance, consider the metric of $\mathbb{R}^4$ parametrized as

$$ds^2 = d\tau^2 + dr^2 + r^2 (d\vartheta^2 + \sin^2 \vartheta d\varphi^2) , \quad (38)$$

where τ is the Euclidean time. The (anti-)self-dual dyonic field that solves the Maxwell equations by virtue of Bianchi identity, i.e. it satisfies $F_{\mu\nu} = \pm \tilde{F}_{\mu\nu}$, is described by the gauge potential

$$A = \mp p \left(\frac{1}{r} d\tau \pm \cos \vartheta d\varphi \right) \quad \rightarrow \quad F = dA = \pm p \left(-\frac{1}{r^2} d\tau \wedge dr \pm \sin \vartheta d\vartheta \wedge d\varphi \right) . \quad (39)$$

This potential, however, is singular at $r = 0$. Additionally, its dual vector field is also singular at $\vartheta = 0, \pi$; this singularity can be removed by performing a gauge transformation, as in the case of the magnetic monopole. Nevertheless, when gravity is included, the singularity can be hidden by an event horizon, producing no violation of the cosmic censorship conjecture whatsoever.

Exercises III

1. Show that the Lagrangian (15) can be written in differential forms as

$$I_M[A] = \frac{1}{2} \int_{\mathcal{M}} F \wedge \star F \quad \text{and} \quad F = dA, \quad (40)$$

is the $U(1)$ field strength 2-form and $\star : \Omega^p(\mathcal{M}) \rightarrow \Omega^{D-p}(\mathcal{M})$ is the Hodge dual, defined as acting on the bases of differential forms as

$$\star(dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p}) = \frac{\sqrt{|g|}}{(D-p)!} \varepsilon^{\mu_1 \dots \mu_p}_{\mu_{p+1} \dots \mu_D} dx^{\mu_{p+1}} \wedge \dots \wedge dx^{\mu_D}. \quad (41)$$

2. Show that Eq. (17) can be written as Eq. (31).
3. Show that the electric and magnetic charges can be respectively defined as

$$Q = \oint_{\mathbb{S}^2} \star F \quad \text{and} \quad \tilde{Q} = \oint_{\mathbb{S}^2} F. \quad (42)$$

IV. SU(2) YANG-MILLS THEORY

In the standard model of particle physics, weak interactions are described by $SU(2)$ gauge theories, with fermions and gauge bosons in the fundamental and adjoint representation, respectively. In this section, we will focus on the bosonic sector, focusing on topologically nontrivial configurations characterized by its Chern-Pontryagin index.

In these notes, we use $t_i = -\frac{i}{2}\tau_i$ as the anti-Hermitian generators of $SU(2)$, where τ_i are the Pauli matrices, and Latin characters denote $SU(2)$ indices. These generators satisfy the $SU(2)$ algebra, i.e. $[t_i, t_j] = \epsilon_{ijk} t^k$ and, using the anticommutation relation $\{t_i, t_j\} = 2\delta_{ij}\mathbb{I}$, one obtains

$$t_i t_j = -\frac{1}{4}\delta_{ij}\mathbb{I} + \frac{1}{2}\epsilon_{ijk} t^k. \quad (43)$$

The latter implies that $\text{Tr}(t_i t_j) = -\frac{1}{2}\delta_{ij}$ and $\text{Tr}(t_i t_j t_k) = -\frac{1}{4}\epsilon_{ijk}$. It is convenient to introduce the

left-invariant forms of $SU(2)$, which can be parametrized in terms of the Euler angles as

$$\sigma_1 = \cos \psi d\vartheta + \sin \vartheta \sin \psi d\varphi, \quad (44a)$$

$$\sigma_2 = -\sin \psi d\vartheta + \sin \vartheta \cos \psi d\varphi, \quad (44b)$$

$$\sigma_3 = d\psi + \cos \vartheta d\varphi. \quad (44c)$$

These 1-forms satisfy the Maurer-Cartan equation $d\sigma_i + \frac{1}{2}\epsilon_{ijk}\sigma^j \wedge \sigma^k = 0$, where $SU(2)$ indices are raised and lowered by the Cartan-Killing metric of such a group, and $\wedge$ denotes the wedge product of differential forms. The line element of the round two-sphere and, using the Hopf fibration, that of the three-sphere can be respectively written in terms of the Maurer-Cartan 1-forms as

$$d\Omega_{(2)}^2 = \sigma_2^2 + \sigma_3^2 = d\vartheta^2 + \sin^2 \vartheta d\varphi^2, \quad (45a)$$

$$d\Omega_{(3)}^2 = \frac{1}{4} (\sigma_1^2 + \sigma_2^2 + \sigma_3^2) = \frac{1}{4} (d\vartheta^2 + \sin^2 \vartheta d\varphi^2 + (d\psi + \cos \vartheta d\varphi)^2). \quad (45b)$$

The field strength of the non-Abelian gauge connection is defined in Eq. (8). In the case of $SU(2)$ Yang-Mills field, it can be explicitly written as

$$A = A_\mu^i t_i dx^\mu. \quad (46)$$

In a Riemannian manifold $(\mathcal{M}, g)$, the dynamics of the Yang-Mills theory for the $SU(2)$ group is dictated by the action principle

$$I_{\text{YM}} = -\frac{1}{2} \int_{\mathcal{M}} d^4x \sqrt{|g|} \text{Tr}(F_{\mu\nu} F^{\mu\nu}), \quad (47)$$

where the trace is taken over the $SU(2)$ indices. The field equations are obtained by performing stationary variations of this action with respect to A_μ , giving

$$\nabla_\mu F^{\mu\nu} + [A_\mu, F^{\mu\nu}] = 0. \quad (48)$$

This is a second-order equation for the non-Abelian gauge field, which, in general, is difficult to solve. The reason is the nonlinear term arising from the last commutator; the latter is absent in Maxwell equations (16). However, it is possible to reduce the order of these equations by imposing (anti-)self-duality, significantly simplifying the nonlinear system. Additionally, the stress-energy tensor can be obtained by performing arbitrary variations of the Yang-Mills action with respect to the metric, giving

$$T_{\mu\nu} = -2 \text{Tr} \left(F_{\mu\lambda} F_\nu{}^\lambda - \frac{1}{4} g_{\mu\nu} F_{\lambda\rho} F^{\lambda\rho} \right) = -\text{Tr} \left(g^{\lambda\rho} [F_{\mu\lambda} + \tilde{F}_{\mu\lambda}] [F_{\nu\rho} - \tilde{F}_{\nu\rho}] \right). \quad (49)$$

Like the Abelian case, it vanishes identically for (anti-)self-dual configurations, although the field can be nontrivial.

A. Instantons

Instantons are (anti-)self-dual stationary solutions of Euclidean Yang-Mills theory that characterize the ground state of homotopically inequivalent vacua. They are asymptotically pure-gauge configurations representing topologically nontrivial maps between the boundary and the group manifold. In four dimensions, they are characterized by the $SU(2)$ Chern-Pontryagin index, i.e.

$$C_2[A] = -\frac{1}{8\pi^2} \int_{\mathcal{M}} d^4x \sqrt{|g|} \operatorname{Tr} \left(\tilde{F}_{\mu\nu} F^{\mu\nu} \right). \quad (50)$$

This is a topological invariant of the second Pontryagin class, which does not depend on the Riemannian metric on $\mathcal{M}$. It is an integer that measures the number of instantons and is related to the difference between the number of positive and negative chiral spinors. Similar to the Abelian case, its addition to the Yang-Mills action does not modify the equations of motion, but it changes the action and variations thereof in a nontrivial way. Indeed, as shown in Ref. [21, 38], the presence of instantons introduces the Chern-Pontryagin term at the level of the effective action, whose physical consequences have been widely studied.

One of the most important instanton solutions in $SU(2)$ Yang-Mills theory is the Belavin-Polyakov-Schwartz-Tyupkin pseudoparticle configuration on $\mathbb{R}^4$ [39]. It can be obtained by assuming the line element of the four-dimensional Euclidean flat space and a non-Abelian gauge field parametrized as

$$ds^2 = dr^2 + \frac{r^2}{4}(\sigma_1^2 + \sigma_2^2 + \sigma_3^2) \quad \text{and} \quad A = a(r) \sigma^i t_i. \quad (51)$$

As previously mentioned, the Yang-Mills Eqs. (48) are automatically satisfied by the (anti-)self-dual condition $F_{\mu\nu} = \pm \tilde{F}_{\mu\nu}$. Replacing these ansätze into the latter, we obtain the first-order equation

$$r a' \pm 2a(1 - a) = 0, \quad (52)$$

where $a = a(r)$ and prime denotes differentiation with respect to the radial coordinate. Depending

on the sign, there are two possible solutions to this equation; for the $\pm$, they are

$$a(r) = \frac{\lambda^2}{r^2 + \lambda^2} \quad \text{and} \quad a(r) = \frac{r^2}{r^2 + \lambda^2}, \quad (53)$$

respectively, where λ is an integration constant dubbed the instanton size. These solutions are regular $\forall r \in \mathbb{R}_{\geq 0}$ as it can be checked from its gauge invariants

$$\text{Tr}(F_{\mu\nu}F^{\mu\nu}) = \pm \text{Tr}(\tilde{F}_{\mu\nu}F^{\mu\nu}) = -\frac{96\lambda^4}{(r^2 + \lambda^2)^4}. \quad (54)$$

Additionally, the self-dual solution is asymptotically trivial, as $a(r) \rightarrow 0$ when $r \rightarrow \infty$, while the anti-self-dual behaves as $a(r) \rightarrow 1$ when $r \rightarrow \infty$, being asymptotically pure gauge; the latter can be checked by using the Maurer-Cartan equation for the left-invariant forms of $SU(2)$. Direct evaluation of the Yang-Mills action (47) and Chern-Pontryagin index (50) on these (anti-)self-dual instantons gives

$$I_{\text{YM}} = 8\pi^2 \quad \text{and} \quad C_2[A] = \pm 1, \quad (55)$$

respectively, being BPS states with nontrivial topological charge. One can check that the stress-energy tensor in Eq. (49) vanishes identically when evaluated on these solutions.

B. Merons

Besides instantons, merons — first found by de Alfaro, Fubini, and Furlan in Ref. [40] — represent simple yet nontrivial solutions to Yang-Mills equations with fractional topological charge. This implies they must be singular on $\mathbb{R}^4$ and cannot be isolated in nature: it has been proposed that instantons can be thought of as a bound state of two merons in the limit when the instanton size goes to zero. Merons' singularity can be hidden behind an event horizon when their backreaction is considered [41, 42], and they can be used to construct Euclidean AdS wormholes connecting two different asymptotic regions [43].

Merons are field configurations described by a gauge potential proportional to a pure-gauge field. In general, the nonlinear nature of Yang-Mills equations implies that the field strength is not necessarily vanishing if this condition is satisfied, since

$$A = \lambda U^{-1} dU \quad \rightarrow \quad F = \lambda(1 - \lambda) dU^{-1} \wedge dU. \quad (56)$$

Thus, it is clear that the non-Abelian field strength vanishes for trivial ($\lambda = 0$) or pure-gauge configurations ($\lambda = 1$), but it does not vanish otherwise. Indeed, as shown in Ref. [40], a $SU(2)$ configuration that solves the Yang-Mills equations (48) in four-dimensional Euclidean space is given by

$$ds^2 = dr^2 + \frac{r^2}{4}(\sigma_1^2 + \sigma_2^2 + \sigma_3^2) \quad \text{and} \quad A = \frac{1}{2} \sigma^i t_i, \quad (57)$$

which corresponds to the ansatz (56) with $U^{-1}dU = \sigma^i t_i$ with $\lambda = 1/2$. This configuration is not (anti-)self-dual. Additionally, noticing that the Pontryagin density can be written as a total derivative via $\text{Tr}(F \wedge F) = dC_P$ where C_P is the Chern-Simons form (see exercise IV. 1), one can compute the meronic topological charge via

$$I_{\text{CS}} = \frac{1}{8\pi^2} \int_{\mathbb{S}^3} \text{Tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right) = \frac{1}{2}. \quad (58)$$

The fractional value of the topological charge suggests that $SU(2)$ merons are singular. This can be checked by computing the gauge invariant

$$\text{Tr}(F_{\mu\nu} F^{\mu\nu}) = -\frac{3}{r^4}, \quad (59)$$

which is divergent if $r = 0$. Additionally, notice that $A_{(\text{inst.})} \approx A_{(\text{meron})} + \mathcal{O}(\lambda^2)$ if $\lambda \ll 1$; this is why instantons are usually thought of as a bound state of two merons in the small-instanton size approximation.

Exercises IV

1. Using differential forms, show that the Pontryagin density can be written as a boundary term, i.e.

$$\frac{1}{2} d^4x \sqrt{|g|} \text{Tr}(\tilde{F}_{\mu\nu} F^{\mu\nu}) = \text{Tr}(F \wedge F) = dC_P, \quad (60)$$

where the Chern-Simons form is defined as

$$C_P = \text{Tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right) = \text{Tr} \left(A \wedge F - \frac{1}{3} A \wedge A \wedge A \right). \quad (61)$$

2. Check Eqs. (55) and (58) by explicit calculation.

V. AXIAL ANOMALY

Anomalies in quantum field theories are important phenomena that occur when a classical symmetry is broken at the quantum level. Their presence is closely related to the nontrivial topology of fiber bundles, and they represent one of the most fascinating connections between mathematics and physics. This section reviews the basics of a particular one, the axial (or chiral) anomaly. Although the topic is technical from a mathematical viewpoint, we try to keep the discussion as simple as possible.

Let us start by considering the Dirac action for a massless spinor in a four-dimensional Minkowski space, that is,

$$I_D[\bar{\Psi}, \Psi] = - \int_{\mathcal{M}_4} d^4x \bar{\Psi} \not{\partial} \Psi := \int_{\mathcal{M}_4} d^4x \mathcal{L}_D, \quad (62)$$

where $\mathcal{L}_D$ is the Dirac Lagrangian and we are using the Feynmann notation $\not{\partial} = \gamma^\mu \partial_\mu$, with γ_μ being the Dirac gamma matrices satisfying the Clifford algebra $\{\gamma_\mu, \gamma_\nu\} = 2\eta_{\mu\nu}\mathbb{I}$, and $\bar{\Psi} = i\Psi^\dagger \gamma^0$ is the adjoint Dirac spinor. Invariance under global Lorentz transformations in the spinorial representation implies that Hermiticity properties of gamma matrices can be summarized as $(\gamma^\mu)^\dagger = \gamma^0 \gamma^\mu \gamma^0$, namely, they are Hermitian for $\mu = 1, 2, 3$ and anti-Hermitian for $\mu = 0$. Additionally, one can define $\gamma_5 = i\gamma_0\gamma_1\gamma_2\gamma_3$ as the chiral gamma matrix, satisfying $(\gamma_5)^\dagger = \gamma_5$ and the anticommutation relation $\{\gamma_\mu, \gamma_5\} = 0$.

The massless Dirac action is invariant under global $U(1)$ vector and axial transformations, i.e.,

$$U(1)_V : \quad \Psi(x) \rightarrow \Psi'(x) = e^{i\alpha} \Psi(x) \quad \text{and} \quad \bar{\Psi}(x) \rightarrow \bar{\Psi}'(x) = \bar{\Psi}(x) e^{-i\alpha}, \quad (63a)$$

$$U(1)_A : \quad \Psi(x) \rightarrow \Psi'(x) = e^{i\beta\gamma_5} \Psi(x) \quad \text{and} \quad \bar{\Psi}(x) \rightarrow \bar{\Psi}'(x) = \bar{\Psi}(x) e^{i\beta\gamma_5}. \quad (63b)$$

The Noether theorem associated with these symmetries implies the conservation laws

$$\partial_\mu j^\mu(x) = 0 \quad \text{and} \quad \partial_\mu j_5^\mu(x) = 0, \quad (64)$$

where $j^\mu(x) = i\bar{\Psi}(x)\gamma^\mu\Psi(x)$ and $j_5^\mu(x) = i\bar{\Psi}(x)\gamma^\mu\gamma_5\Psi(x)$ are the vector and axial currents, respectively. The existence of a conserved current leads to a conserved charge defined by Eq. (21), which is independent of the choice of the hypersurface Σ . A natural choice is a Cauchy slice of constant

time, say Σ_t . Its induced metric is $h_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu$ with a timelike unit normal normalized such that $n \cdot n = -1$. Then, the conserved charge associated to $U(1)_V$ and $U(1)_A$ can be defined as

$$Q_V = \int_{\Sigma_t} d^3x \sqrt{|h|} n_0 j^0 \quad \text{and} \quad Q_A = \int_{\Sigma_t} d^3x \sqrt{|h|} n_0 j_5^0. \quad (65)$$

In the Weyl basis, the chiral gamma matrix γ_5 and the Dirac spinors can be represented as

$$\gamma_5 = \begin{pmatrix} \mathbb{I} & 0 \\ 0 & -\mathbb{I} \end{pmatrix} \quad \text{and} \quad \Psi(x) = \begin{pmatrix} \psi_+(x) \\ \psi_-(x) \end{pmatrix}, \quad (66)$$

respectively, where $\psi_\pm$ denotes two-column Weyl spinors. Using the projectors $P_\pm = \frac{\mathbb{I} \pm \gamma_5}{2}$, one can define the chiral spinors $\Psi_+ = (\psi_+ \ 0)$ and $\Psi_- = (0 \ \psi_-)$ by projecting the Weyl spinors onto their chiral components, that is, $P_\pm \Psi = \Psi_\pm$. The latter are eigenstates of the chiral gamma matrix with eigenvalues ± 1 , namely, $\gamma_5 \Psi_\pm = \pm \Psi_\pm$. Then, using the Weyl representation in Eq. (66), the temporal components of the vectorial and axial currents are

$$j^0 = i \bar{\Psi} \gamma^0 \Psi = i^2 \Psi^\dagger \gamma^0 \gamma^0 \Psi = \Psi^\dagger \Psi = \psi_+^\dagger \psi_+ + \psi_-^\dagger \psi_-, \quad (67)$$

$$j_5^0 = i \bar{\Psi} \gamma^0 \gamma_5 \Psi = i^2 \Psi^\dagger \gamma^0 \gamma^0 \gamma_5 \Psi = \Psi^\dagger \gamma_5 \Psi = \psi_+^\dagger \psi_+ - \psi_-^\dagger \psi_-, \quad (68)$$

respectively, where we have used that $(\gamma^0)^2 = -\mathbb{I}$. In quantum theory, the classical fields become operators in a Fock space. Expanding the field operators in Fourier modes, we notice that $\psi_\pm^\dagger \psi_\pm$ is related to the density number operator of Weyl spinors with definite chirality. Thus, when acting on a particular state with $n_\pm$ chiral spinors with chirality ± 1 , the vector and axial charge operators in Eq. (65) yield

$$Q_V = n_+ + n_- \quad \text{and} \quad Q_A = n_+ - n_-, \quad (69)$$

respectively. Thus, the conservation of the vectorial current implies that the total electric charge given by the sum of spinors with different chiralities is conserved. However, the conservation of the axial current implies that the difference between the number of positive and negative chiral spinors is conserved. However, the latter is no longer true at the quantum level because of anomalies.

A. Path integral and global $U(1)_V$ transformations

In the following, we will work on the Euclidean section by performing the analytic continuation $t \rightarrow -i\tau$ and $\gamma^\mu \rightarrow i\gamma_E^\mu$, which implies that γ_5 remains invariant. Here, the subscript E in the γ

matrix stands for Euclidean, although we will omit it in what follows. The Clifford algebra for Euclidean gamma matrices is $\{\gamma^\mu, \gamma^\nu\} = -2\delta^{\mu\nu}$ and $\not{\partial}$ is a Hermitian operator.

Before going into the non-perturbative approach to the axial anomaly, let us introduce the path integral for Dirac spinors in Euclidean space and derive the conservation law for the global $U(1)_V$ symmetry. To this end, we consider

$$\mathcal{Z} = \int \mathcal{D}\bar{\Psi} \mathcal{D}\Psi \exp \left(- I_D^{(E)}[\bar{\Psi}, \Psi] \right) = \int \mathcal{D}\bar{\Psi} \mathcal{D}\Psi \exp \left(- \int_{\mathcal{M}_4} d^4x \bar{\Psi} \not{\partial} \Psi \right), \quad (70)$$

where $I_D^{(E)}$ is the analytic continuation of the Dirac action (62) to the Euclidean section, and integration over all trajectories in phase space is taken. To obtain a meaningful quantity from the fermionic measure, let us consider a complete set of eigenstates of the Dirac operator $\{\Psi_n\}$ with eigenvalues $\lambda_n \in \mathbb{R}$, that is, $\not{\partial}\Psi_n = \lambda_n \Psi_n$. Then, we can span the Dirac spinors in terms of these eigenstates as

$$\Psi(x) = \sum_n a_n \Psi_n(x) = \sum_n a_n \langle x|n\rangle \quad \text{and} \quad \bar{\Psi}(x) = \sum_n \Psi_n^\dagger(x) \bar{b}_n = \sum_n \langle n|x\rangle \bar{b}_n, \quad (71)$$

where a_n and $\bar{b}_n$ are Grassmann variables, while $\Psi_n(x)$ are complex functions with $\Psi_n^\dagger(x)$ its transpose conjugate. Orthogonality of the eigenstates implies

$$\int d^4x \Psi_n^\dagger(x) \Psi_m(x) = \delta_{mn} \quad \text{and} \quad \sum_n \Psi_n^\dagger(x) \Psi_n(y) = \delta(x-y). \quad (72)$$

Then, in terms of the eigenstates of $\not{\partial}$ and using the orthogonality conditions, the fermionic measure simplifies, and it can be written in terms of a_n and b_n as

$$\mathcal{D}\bar{\Psi} \mathcal{D}\Psi = (\det \Psi_n^\dagger(x) \det \Psi_m(x))^{-1} \prod_n d\bar{b}_n \prod_m da_m = \prod_n d\bar{b}_n \prod_m da_m, \quad (73)$$

where the Jacobian appears to a negative power because the field transformation is performed over Grassmann variables. Additionally, in the last equality, we have used the property that the determinant of a product of matrices is the product of their determinants, and the inner product

$$(\Psi_n(x), \Psi_m(x)) = \int d^4x \Psi_n^\dagger(x) \Psi_m(x) = \int d^4x \langle n|x\rangle \langle x|m\rangle = \delta_{mn} \quad (74)$$

An important property of the path integral is that, since we are integrating over all configurations, it remains invariant under field redefinitions $\Psi \rightarrow \Psi' = A(x)\Psi$. A particular type of

the latter are *local* $U(1)_V$ transformations, whose element in the exponential representation is $A(x) = U(x) = e^{i\alpha(x)}$. This can be used to derive the first conservation law of Eq. (64) directly. For instance, an infinitesimal transformation $U(1)_V$ transformation reads

$$U(x) = e^{i\alpha(x)} = \mathbb{I} + i\alpha(x) + \mathcal{O}(\alpha^2). \quad (75)$$

Then, using the fact that the path integral remains invariant under field redefinitions, one finds that the relation $\mathcal{Z}[A(x)] = \mathcal{Z}[\mathbb{I}]$ must hold $\forall A(x)$; in particular, for $A(x) = U(x) \in U(1)_V$. Thus, expanding the left-hand side when $\alpha(x) \ll 1$, we find

$$\mathcal{Z}[U(x)] = \mathcal{Z}[\mathbb{I}] + \int d^4x \alpha(x) \frac{\delta}{\delta\alpha(x)} \mathcal{Z}[U(x)] \Big|_{\alpha(x)=0} + \mathcal{O}(\alpha^2). \quad (76)$$

Using this expression, one concludes that, for an arbitrary $\alpha(x)$, we have

$$\frac{\delta}{\delta\alpha(x)} \mathcal{Z}[U(x)] \Big|_{\alpha(x)=0} = 0. \quad (77)$$

Under a local $U(1)_V$ transformation, the Dirac spinors, written in terms of the eigenstates of $\not{\partial}$, transform as

$$\Psi(x) \rightarrow \Psi'(x) = e^{i\alpha(x)} \Psi(x) = (1 + i\alpha(x)) \sum_n a_n \Psi_n(x) := \sum_n a'_n \Psi_n(x), \quad (78a)$$

$$\bar{\Psi}(x) \rightarrow \bar{\Psi}'(x) = \bar{\Psi}(x) e^{-i\alpha(x)} = \sum_n \Psi_n^\dagger(x) \bar{b}_n (1 - i\alpha(x)) := \sum_n \Psi_n^\dagger(x) \bar{b}'_n, \quad (78b)$$

where a'_n and $\bar{b}'_n$ can be obtained from the orthogonality conditions in Eq. (72), giving

$$a'_n = \sum_m C_{mn} a_n, \quad \bar{b}'_n = \sum_m (C_{mn})^\dagger \bar{b}_m, \quad \text{where} \quad C_{mn} = \delta_{mn} + i \int d^4x \alpha(x) \Psi_m^\dagger(x) \Psi_n(x), \quad (79)$$

is a unitary matrix, that is, $(C_{mn})^\dagger = (C_{mn})^{-1}$. Then, the fermionic measure remains invariant under local $U(1)_V$ transformations as the Jacobian is 1, since

$$\mathcal{D}\bar{\Psi} \mathcal{D}\Psi \rightarrow \mathcal{D}\bar{\Psi}' \mathcal{D}\Psi' = \left[\det \sum_l (C_{ln})^\dagger C_{lm} \right]^{-1} \prod_n d\bar{b}_n \prod_m da_m = \prod_n d\bar{b}_n \prod_m da_m. \quad (80)$$

The Dirac action (62), however, is not invariant under local $U(1)_V$ transformations, since its Lagrangian in Euclidean signature transforms as

$$\mathcal{L}_D \rightarrow \mathcal{L}'_D = \bar{\Psi}' \not{\partial} \Psi' = \bar{\Psi} e^{-i\alpha(x)} \gamma^\mu \partial_\mu (e^{i\alpha(x)} \Psi) = \mathcal{L}_D + i\bar{\Psi} \gamma^\mu \Psi \partial_\mu \alpha(x) = \mathcal{L}_D + i\alpha(x) \partial_\mu j^\mu, \quad (81)$$

where $\Psi = \Psi(x)$ and we have integrated by part in the last equality; we have assumed that boundary terms vanish upon suitable boundary conditions at spatial infinity. Additionally, as all Dirac-gamma matrices have been rescaled by i when going to Euclidean signature, the vectorial current in Eq. (81) is $j^\mu = -\bar{\Psi}\gamma^\mu\Psi$. Then, Eq. (77) yields

$$\frac{\delta}{\delta\alpha(x)}\mathcal{Z}[U(x)]\Big|_{\alpha(x)=0} = i \int \mathcal{D}\bar{\Psi} \mathcal{D}\Psi \partial_\mu j^\mu \exp\left(-I_D^{(E)}[\bar{\Psi}, \Psi]\right) = 0 \quad \rightarrow \quad \langle\partial_\mu j^\mu\rangle = 0, \quad (82)$$

meaning that the vectorial current is conserved; this implies the conservation of the electric charge. In the case of the $U(1)_A$ symmetry, however, the fermionic measure does not remain invariant, inducing an anomaly at the quantum level. Let us discuss this case in the following.

B. Noninvariance of the fermionic measure under $U(1)_A$ transformations: the anomaly

Fujikawa performed the nonperturbative computation of the axial anomaly in Dirac-Yang-Mills theory using the path integral approach in Refs. [30, 31]. Here, we follow his main steps in the simplest scenario, namely, Dirac-Maxwell theory. To do so, let us consider the path integral for quantum electrodynamics in flat space, neglecting the Faddeev-Popov determinant and the sources for the sake of simplicity, i.e.,

$$\begin{aligned} \mathcal{Z} &= \int \mathcal{D}A \mathcal{D}\bar{\Psi} \mathcal{D}\Psi \exp\left(-I_{\text{QED}}^{(E)}[\bar{\Psi}, \Psi]\right) \\ &= \int \mathcal{D}A \mathcal{D}\bar{\Psi} \mathcal{D}\Psi \exp\left[-\int_{\mathcal{M}_4} d^4x \left(\bar{\Psi}\not{D}\Psi + \frac{1}{4}F_{\mu\nu}F^{\mu\nu}\right)\right], \end{aligned} \quad (83)$$

where $\not{D} = \gamma^\mu D_\mu = \gamma^\mu(\partial_\mu + A_\mu)$ is the Hermitian Abelian covariant derivative in Euclidean signature. The key point in Fujikawa's computation is that, in contrast to vector $U(1)$ transformations, the fermionic measure does not remain invariant under axial $U(1)$ transformations; this is the root of the axial anomaly. To see this, let us expand the measure in terms of eigenstates of the Dirac-Maxwell operator, say $\{\Psi_n\}$, such that $\not{D}\Psi_n = \lambda_n\Psi_n$. From hereon, the eigenstates of $\not{D}$ should not be confused with those of $\not{\partial}$, even though we use the same notation for both. Then, in this case, the Dirac spinors admit a similar expansion in terms of these eigenstates [cf. Eq. (71)] and they follow the same orthogonality conditions as in Eq. (72).

As the path integral remains invariant under field redefinition, we can follow exactly the same procedure as in the $U(1)_V$ case, namely, to promote the global axial symmetry to be a local one

and derive the (non-)conservation law from Eq. (77). In terms of the eigenstates of $\not{D}$, the Dirac spinors transform under $U(1)_A$ transformations as

$$\Psi(x) \rightarrow \Psi'(x) = e^{i\beta(x)\gamma_5} \Psi(x) := \sum_n a''_n \Psi_n(x), \quad (84a)$$

$$\bar{\Psi}(x) \rightarrow \bar{\Psi}'(x) = \bar{\Psi}(x) e^{i\beta(x)\gamma_5} := \sum_n \Psi_n^\dagger(x) \bar{b}''_n, \quad (84b)$$

where the transformed Grassmann variables are related to the previous ones by

$$a''_n = \sum_m B_{mn} a_m, \quad \bar{b}''_n = \sum_m B_{mn} \bar{b}_m, \quad \text{where} \quad B_{mn} = \delta_{mn} + i \int d^4x \beta(x) \Psi_n^\dagger(x) \gamma_5 \Psi_m(x), \quad (85)$$

is also a unitary matrix since γ_5 is Hermitian. However, the transformation law of the Grassmann variables under $U(1)_A$ transformations implies that the fermionic measure transforms as

$$\mathcal{D}\bar{\Psi}\mathcal{D}\Psi \rightarrow \mathcal{D}\bar{\Psi}'\mathcal{D}\Psi' = \left[\det \sum_l B_{ml} B_{ln} \right]^{-1} \prod_n d\bar{b}_n \prod_m da_m = \left[\det B_{mn} \right]^{-2} \prod_n d\bar{b}_n \prod_m da_m. \quad (86)$$

The Jacobian is not necessarily equal to 1 as in the case of $U(1)_V$ transformation of the fermionic measure. In fact, this determinant is formally divergent, and it must be regularized. We will discuss some of the details in the following.

To compute the determinant, we use the well-known formula $\det B = \exp \text{Tr} \ln B$ and expand the logarithm to first order in $\beta(x)$ using $\ln(1 + \epsilon) \approx \epsilon + \mathcal{O}(\epsilon)$, when $\epsilon \ll 1$. Then, the Jacobian can be written as

$$\left[\det B_{mn} \right]^{-2} = \exp \left(-2i \int d^4x \beta(x) \sum_n \Psi_n(x)^\dagger \gamma_5 \Psi(x) \right). \quad (87)$$

The trace in the exponential is divergent by the orthogonality condition (72), as it is proportional to $\delta(0)$. There are different ways of regularizing this divergence, and all of them lead to the same result [44]. Let us focus on the method used by Fujikawa by introducing a gauge invariant cutoff, that is,

$$\begin{aligned} \sum_n \Psi_n^\dagger(x) \gamma_5 \Psi_n(x) &= \lim_{M \rightarrow \infty} \sum_n \Psi_n^\dagger(x) \gamma_5 \exp \left[-\frac{\lambda_n^2}{M^2} \right] \Psi_n(x) \\ &= \lim_{M \rightarrow \infty} \sum_n \Psi_n^\dagger(x) \gamma_5 \exp \left[-\frac{\not{D}^2}{M^2} \right] \Psi_n(x), \end{aligned} \quad (88)$$

where $\not{D}^2 = \gamma^\mu \gamma^\nu D_\mu D_\nu$. This operator can be decomposed in terms of its symmetric and skew-symmetric part as

$$\begin{aligned}\not{D}^2 &= \gamma^\mu \gamma^\nu D_\mu D_\nu = \frac{1}{2} \left(\{\gamma^\mu, \gamma^\nu\} + [\gamma^\mu, \gamma^\nu] \right) D_\mu D_\nu = -D_\mu D^\mu + \gamma^{\mu\nu} D_\mu D_\nu \\ &= -D_\mu D^\mu + \frac{1}{2} \gamma^{\mu\nu} F_{\mu\nu},\end{aligned}\tag{89}$$

where we have defined $\gamma_{\mu_1 \dots \mu_p} = \gamma_{[\mu_1} \dots \gamma_{\mu_p]}$ and the relations $\{\gamma^\mu, \gamma^\nu\} = -2\delta^{\mu\nu} \mathbb{I}$ and $[D_\mu, D_\nu] = F_{\mu\nu}$ have been used. Then, the eigenstates of $\not{D}$ can be written in momentum space by using their Fourier transform, i.e.,

$$\Psi_n(x) = \int \frac{d^4 k}{(2\pi)^2} e^{i k_\mu x^\mu} \tilde{\Psi}_n(k) \quad \text{and} \quad \Psi_n^\dagger(x) = \int \frac{d^4 k}{(2\pi)^2} e^{-i k_\mu x^\mu} \tilde{\Psi}_n^\dagger(k).\tag{90}$$

Replacing these expressions and Eq. (89) into regularized trace in Eq. (88), we obtain

$$\begin{aligned}\sum_n \Psi_n^\dagger(x) \gamma_5 \Psi_n(x) &= \lim_{M \rightarrow \infty} \int \frac{d^4 k}{(2\pi)^2} \frac{d^4 k'}{(2\pi)^2} \sum_n e^{-i k_\mu x^\mu} \tilde{\Psi}_n^\dagger(k) \gamma_5 \exp \left[-\frac{\not{D}^2}{M^2} \right] e^{i k'_\mu x^\mu} \tilde{\Psi}_n(k') \\ &= \lim_{M \rightarrow \infty} \int \frac{d^4 k}{(2\pi)^4} \text{Tr} \left(e^{-i k_\mu x^\mu} \gamma_5 \exp \left[-\frac{\not{D}^2}{M^2} \right] e^{i k_\mu x^\mu} \right) \\ &= \lim_{M \rightarrow \infty} \int \frac{d^4 k}{(2\pi)^4} \text{Tr} \left(\gamma_5 \exp \left[\frac{(D_\mu + i k_\mu)(D^\mu + i k^\mu)}{M^2} - \frac{\gamma^{\mu\nu} F_{\mu\nu}}{2M^2} \right] \right) \\ &= \lim_{M \rightarrow \infty} M^4 \int \frac{d^4 k}{(2\pi)^4} e^{-k_\mu k^\mu} \text{Tr} \left(\gamma_5 \exp \left[\frac{D_\mu D^\mu}{M^2} + \frac{2i k^\mu D_\mu}{M} - \frac{\gamma_\mu \gamma_\nu F^{\mu\nu}}{2M^2} \right] \right).\end{aligned}\tag{91}$$

In the last line, we have rescaled the integration variable as $k^\mu \rightarrow M k^\mu$. In the third line, we have used the fact that $e^{-i k_\mu x^\mu} f(\partial_\mu) e^{i k_\mu x^\mu} = f(\partial_\mu + i k_\mu)$ and, from the second one, Tr denotes trace over Dirac matrices, that is,

$$\sum_n \tilde{\Psi}_n^\dagger(k) \gamma_{\mu_1} \gamma_{\mu_2} \dots \gamma_{\mu_p} \Psi_n(k') = \text{Tr}(\gamma_{\mu_1} \gamma_{\mu_2} \dots \gamma_{\mu_p}) \delta(k - k').\tag{92}$$

To evaluate the integral in Eq. (91), we power expand the exponential. We notice that different traces will appear, all of them involving γ_5 . However, only the piece involving the Maxwell field strength will contribute. This can be seen by noticing that the trace of any odd number of gamma matrices vanishes and

$$\text{Tr}(\gamma_5) = \text{Tr}(\gamma_5 \gamma_\mu) = \text{Tr}(\gamma_5 \gamma_\mu \gamma_\nu) = \text{Tr}(\gamma_5 \gamma_\mu \gamma_\nu \gamma_\lambda) = 0,\tag{93a}$$

$$\text{Tr}(\gamma_5 \gamma_\mu \gamma_\nu \gamma_\lambda \gamma_\rho) = -4 \varepsilon_{\mu\nu\lambda\rho}.\tag{93b}$$

Then, only the third term of Eq. (91) contributes, and the trace becomes

$$\begin{aligned}
\sum_n \Psi_n^\dagger(x) \gamma_5 \Psi_n(x) &= \lim_{M \rightarrow \infty} \frac{1}{2!} \frac{M^4}{4M^4} \int \frac{d^4 k}{(2\pi)^4} e^{-k_\mu k^\mu} \underbrace{\text{Tr}(\gamma_5 \gamma_\mu \gamma_\nu \gamma_\lambda \gamma_\rho)}_{-4 \varepsilon_{\mu\nu\lambda\rho}} F^{\mu\nu} F^{\lambda\rho} \\
&= -\frac{1}{2} \underbrace{\int \frac{d^4 k}{(2\pi)^4} e^{-k_\mu k^\mu}}_{(16\pi)^{-2}} \varepsilon_{\mu\nu\lambda\rho} F^{\mu\nu} F^{\lambda\rho} \\
&= -\frac{1}{32\pi^2} \varepsilon_{\mu\nu\lambda\rho} F^{\mu\nu} F^{\lambda\rho}.
\end{aligned} \tag{94}$$

where Gaussian integration was performed in the last line. Then, using the definition of the dual field strength, i.e. $\tilde{F}_{\mu\nu} = \frac{1}{2} \varepsilon_{\mu\nu\lambda\rho} F^{\lambda\rho}$, we conclude that the regularized trace is

$$\sum_n \Psi_n^\dagger(x) \gamma_5 \Psi_n(x) = -\frac{1}{16\pi^2} \tilde{F}_{\mu\nu} F^{\mu\nu}. \tag{95}$$

Finally, the regularized determinant is

$$\left[\det B_{mn} \right]^{-2} = \exp \left(\frac{i}{8\pi^2} \int d^4 x \beta(x) \tilde{F}_{\mu\nu} F^{\mu\nu} \right). \tag{96}$$

As discussed in the previous Subsection, the path integral remains invariant under field redefinitions, in particular, under a local $U(1)_A$ transformation. The classical Dirac action, however, does not remain invariant as

$$\begin{aligned}
\mathcal{L}_D \rightarrow \mathcal{L}'_D &= \bar{\Psi}' \not{\partial} \Psi' = \bar{\Psi} e^{i\beta(x)\gamma_5} \gamma^\mu \partial_\mu (e^{i\beta(x)\gamma_5} \Psi) = \mathcal{L}_D + i\bar{\Psi} \gamma^\mu \gamma_5 \Psi \partial_\mu \beta(x) \\
&= \mathcal{L}_D + i\beta(x) \partial_\mu j_5^\mu,
\end{aligned} \tag{97}$$

Thus, considering the nontrivial transformation of the fermionic measure under $U(1)_A$ field redefinition as given in Eq. (87), the path integral becomes

$$\mathcal{Z}[U(x)] = \int \mathcal{D}A \mathcal{D}\bar{\Psi} \mathcal{D}\Psi \exp \left[-I_{\text{QED}}^{(\text{E})} + i \int_{\mathcal{M}_4} d^4 x \beta(x) \left(\partial_\mu j_5^\mu + \frac{1}{8\pi^2} \tilde{F}_{\mu\nu} F^{\mu\nu} \right) \right]. \tag{98}$$

Therefore, invariance of the path integral under an axial field redefinition implies that, at the quantum level, the axial current is no longer conserved but, according to Eq. (77), its divergence is proportional to the Pontryagin density of $U(1)$, namely,

$$\langle \partial_\mu j_5^\mu \rangle = -\frac{1}{8\pi^2} \tilde{F}_{\mu\nu} F^{\mu\nu}. \tag{99}$$

This is known as the axial anomaly. Adler-Bell-Jackiw obtained this result using perturbation theory at one loop [14, 15]. Thus, one concludes that the axial anomaly is one-loop exact. Moreover, it is independent of the regularization scheme, as shown in different cases [44]. Indeed, if one considers different connections, even non-Abelian gauge fields and the Lorentz connection of gravity theories, the axial anomaly acquires additional contributions, that is,

$$\langle \partial_\mu j_5^\mu \rangle = -\frac{1}{8\pi^2} \tilde{F}_{\mu\nu} F^{\mu\nu} - \frac{1}{8\pi^2} \text{Tr} [\tilde{G}_{\mu\nu} G^{\mu\nu}] - \frac{1}{384\pi^2} \tilde{R}_{\mu\nu\lambda\rho} R^{\mu\nu\lambda\rho}, \quad (100)$$

where $G_{\mu\nu}$ denotes the non-Abelian field strength and $R_{\mu\nu\lambda\rho}$ the Riemann tensor, while tilde stands for their Hodge duals. Indeed, the integral of this equation is related to the index theorem of Atiyah-Patodi-Singer [45–47], which accounts for the difference between the number of positive and negative chiral spinors, exactly what the (non-)conservation of the chiral charge is related to. Therefore, if the integral of the right-hand side of Eq. (100) is nonvanishing, the chiral charge will not be conserved; this is exactly what happens with instantons and with other topologically nontrivial configurations in gauge theory and gravity.

VI. GENERAL RELATIVITY

As Euclidean general relativity in four dimensions can be formulated as a gauge theory for the $SO(4)$ group, it is natural to ask whether it hosts gravitational instantons as part of its solution space. If so, they would be relevant in understanding the global properties of the quantum-gravity vacuum, and they will contribute to anomalies when quantum fields are considered on these backgrounds. As we will study in this section, there exist interesting geometries that belong to the space of solutions of general relativity that resemble instantons in Yang-Mills theories. They will be dubbed gravitational instantons.

A. Einstein-AdS gravity

In this section, we consider Einstein's theory of general relativity in Euclidean signature with a negative cosmological constant, i.e., $\Lambda = -3/\ell^2$, where ℓ is the AdS radius. The dynamics of the theory is dictated by the action principle

$$I_{\text{EH}}[g_{\mu\nu}] = \kappa \int_{\mathcal{M}} d^4x \sqrt{|g|} \left(R + \frac{6}{\ell^2} \right) + I_{\text{bdy}}, \quad (101)$$

where $\kappa = (16\pi G)^{-1}$ is the gravitational constant, G is the Newton's constant, $g = \det g_{\mu\nu}$ is the metric determinant, and $R = g^{\mu\nu} R_{\mu\nu}$ is the Ricci scalar, with $R_{\mu\nu} = R^\lambda_{\mu\lambda\nu}$ and $R^\lambda_{\mu\rho\nu}$ being the Ricci and Riemann tensor, respectively. Additionally, it is convenient to define the displaced Riemann curvature as

$$F_{\lambda\rho}^{\mu\nu} := R_{\lambda\rho}^{\mu\nu} + \frac{1}{\ell^2} \delta_{\lambda\rho}^{\mu\nu}, \quad (102)$$

that measures the deviation from a maximally symmetric space with negative-constant curvature, with $\delta_{\nu_1 \dots \nu_p}^{\mu_1 \dots \mu_p} = p! \delta_{[\nu_1}^{[\mu_1} \dots \delta_{\nu_p]}^{\mu_p]}$ being the generalized Kronecker delta of order p , satisfying

$$\delta_{\nu_1 \dots \nu_p}^{\mu_1 \dots \mu_p} \delta_{\mu_1}^{\nu_1} \dots \delta_{\mu_k}^{\nu_k} = \frac{(D-p+k)!}{(D-p)!} \delta_{\nu_{k+1} \dots \nu_p}^{\mu_{k+1} \dots \mu_p}, \quad (103)$$

where $D = \dim \mathcal{M}$ and $k \leq p \leq D$. The boundary term I_{bdy} , on the other hand, is included to render the variational principle well posed without affecting the bulk dynamics of the metric, and it cancels the divergencies arising in asymptotically locally AdS spacetimes.

The field equations can be obtained by performing arbitrary variations of the action (101) with respect to the metric, giving $\mathcal{E}_\nu^\mu = 0$, where

$$\mathcal{E}_\nu^\mu := R_\nu^\mu - \frac{1}{2} \delta_\nu^\mu R - \frac{3}{\ell^2} \delta_\nu^\mu = -\frac{1}{4} \delta_{\nu\lambda\rho}^{\mu\alpha\beta} \left(R_{\alpha\beta}^{\lambda\rho} + \frac{1}{\ell^2} \delta_{\alpha\beta}^{\lambda\rho} \right) = -\frac{1}{4} \delta_{\nu\lambda\rho}^{\mu\alpha\beta} F_{\alpha\beta}^{\lambda\rho} = 0. \quad (104)$$

Taking the trace on the last equation, one obtains $R = -\frac{12}{\ell^2}$. Then, inserting this result back into Eq. (104), one obtains the trace-reversed Einstein's equations; they are

$$R_\nu^\mu = -\frac{3}{\ell^2} \delta_\nu^\mu. \quad (105)$$

In four dimensions, all Einstein-AdS spaces are characterized by the condition (105). This, in turn, implies that Weyl and the Riemann tensors are related simply. To see this, consider the decomposition of the Riemann tensor into its irreducible pieces, that is,

$$R_{\lambda\rho}^{\mu\nu} = W_{\lambda\rho}^{\mu\nu} + 2 \delta_{[\lambda}^{[\mu} H_{\rho]}^{\nu]} + \frac{1}{12} \delta_{\lambda\rho}^{\mu\nu} R, \quad (106)$$

where $W_{\lambda\rho}^{\mu\nu}$ is the Weyl tensor, which is traceless under contraction of any of their indices, the traceless Ricci tensor is defined as $H_\nu^\mu = R_\nu^\mu - \frac{1}{4} \delta_\nu^\mu R$. If equation (105) is satisfied, then $H_\nu^\mu = 0$. This occurs always for Einstein spaces. In this sense, the traceless Ricci tensor measures deviations from the Einstein condition (105). Additionally, using the fact that $R = -\frac{12}{\ell^2}$, one concludes that, for Einstein spaces, the Weyl tensor becomes

$$W_{\lambda\rho}^{\mu\nu}|_{\text{E}} = R_{\lambda\rho}^{\mu\nu} + \frac{1}{\ell^2} \delta_{\lambda\rho}^{\mu\nu} = F_{\lambda\rho}^{\mu\nu}. \quad (107)$$

B. Topological renormalization

There are different possibilities for choosing the boundary term I_{bdy} in Eq. (101). Here, we will focus on the Chern form as a counterterm; this prescription is also known as Kounterterms [7, 48–50], and it coincides with intrinsic boundary counterterms when the Fefferman-Graham expansion is considered [51]. In four dimensions, the codimension-1 integral of the Chern form is related to the bulk integral of the Gauss-Bonnet term via

$$\int_{\mathcal{M}} d^4x \sqrt{|g|} \mathcal{G} = 32\pi^2 \chi(\mathcal{M}) + 4 \int_{\partial\mathcal{M}} d^3x \sqrt{|h|} \delta_{\lambda\rho\beta}^{\mu\nu\alpha} K_{\mu}^{\lambda} \left(\frac{1}{2} \mathcal{R}_{\nu\alpha}^{\rho\beta} - \frac{1}{3} K_{\nu}^{\rho} K_{\alpha}^{\beta} \right), \quad (108)$$

where $\chi(\mathcal{M})$ is the Euler characteristic and

$$\mathcal{G} = \frac{1}{4} \delta_{\nu_1 \dots \nu_4}^{\mu_1 \dots \mu_4} R_{\mu_1 \mu_2}^{\nu_1 \nu_2} R_{\mu_3 \mu_4}^{\nu_3 \nu_4} = R^2 - 4R_{\mu\nu} R^{\mu\nu} + R_{\mu\nu\lambda\rho} R^{\mu\nu\lambda\rho}, \quad (109)$$

is the Gauss-Bonnet term. Additionally, $h_{\mu\nu} = g_{\mu\nu} - n_{\mu} n_{\nu}$ is the induced metric on $\partial\mathcal{M}$ with $n = n^{\mu} \partial_{\mu}$ being its space-like unit normal satisfying $n_{\mu} n^{\mu} = 1$. Additionally, $\mathcal{R}_{\lambda\rho}^{\mu\nu} = \mathcal{R}_{\lambda\rho}^{\mu\nu}(h)$ and $K_{\mu\nu} = h_{\mu}^{\lambda} \nabla_{\lambda} n_{\nu}$ are the intrinsic and extrinsic curvature, respectively; the trace of the latter is denoted by $K = g^{\mu\nu} K_{\mu\nu}$. These quantities are related through the Gauss-Codazzi relation

$$\mathcal{R}_{\lambda\rho}^{\mu\nu} = h_{\alpha}^{\mu} h_{\beta}^{\nu} h_{\lambda}^{\sigma} h_{\rho}^{\eta} R_{\sigma\eta}^{\alpha\beta} + 2K_{[\lambda}^{\mu} K_{\rho]}^{\nu}. \quad (110)$$

The addition of the Gauss-Bonnet term to the Einstein-AdS action or, equivalently, the Chern form by virtue of the Gauss-Bonnet theorem (108), provides a suitable setup for renormalizing the action and variations thereof in asymptotically locally AdS spacetimes. Specifically, adding the Gauss-Bonnet term with an arbitrary coupling α , i.e.,

$$I_{\text{EH}}^{(\text{ren})} = \kappa \int_{\mathcal{M}} d^4x \sqrt{|g|} \left(R + \frac{6}{\ell^2} + \alpha \mathcal{G} \right), \quad (111)$$

leads exactly to the same bulk dynamics of Eq. (104). However, the Gauss-Bonnet does change the value of the action and conserved charges. Even more, it becomes divergent in asymptotically locally AdS spacetimes and, therefore, can be used as a counterterm for canceling divergencies. Indeed, renormalization demands a precise coupling α , that is, (see Refs. [7, 48–50], for details)

$$\alpha = \frac{\ell^2}{4}. \quad (112)$$

Then, using this value and the generalized Kronecker delta allows us to rewrite the action in a MacDowell-Mansouri form, namely,

$$\begin{aligned} I_{\text{EH}}^{(\text{ren})} &= \kappa \int_{\mathcal{M}} d^4x \sqrt{|g|} \left(R + \frac{6}{\ell^2} + \frac{\ell^2}{4} \mathcal{G} \right) \\ &= \frac{\kappa \ell^2}{16} \int_{\mathcal{M}} d^4x \sqrt{|g|} \delta_{\nu_1 \dots \nu_4}^{\mu_1 \dots \mu_4} \left(R_{\mu_1 \mu_2}^{\nu_1 \nu_2} + \frac{1}{\ell^2} \delta_{\mu_1 \mu_2}^{\nu_1 \nu_2} \right) \left(R_{\mu_3 \mu_4}^{\nu_3 \nu_4} + \frac{1}{\ell^2} \delta_{\mu_3 \mu_4}^{\nu_3 \nu_4} \right), \end{aligned} \quad (113)$$

This is called the topologically renormalized Einstein-AdS action and, using the Fefferman-Graham expansion, it is equivalent to holographic renormalization [7]. Indeed, it represents the MacDowell-Mansouri action for AdS gravity [52]. Then, evaluating the topologically renormalized action (113) on Einstein-AdS spaces, one concludes that it becomes that of conformal gravity, i.e.,

$$I_{\text{EH}}^{(\text{ren})}|_{\text{E}} = \frac{\kappa \ell^2}{16} \int_{\mathcal{M}} d^4x \sqrt{|g|} \delta_{\nu_1 \dots \nu_4}^{\mu_1 \dots \mu_4} W_{\mu_1 \mu_2}^{\nu_1 \nu_2} W_{\mu_3 \mu_4}^{\nu_3 \nu_4}|_{\text{E}} = \frac{\kappa \ell^2}{4} \int_{\mathcal{M}} d^4x \sqrt{|g|} W_{\lambda \rho}^{\mu \nu} W_{\mu \nu}^{\lambda \rho}|_{\text{E}}, \quad (114)$$

which has been proved to be finite for any asymptotically locally AdS spacetimes, even with relaxed boundary conditions [53].

There is another topological term that can be constructed in four dimensions, known as the Chern-Pontryagin index of $SO(4)$, that is,

$$P_1[\mathcal{M}] = \frac{1}{16\pi^2} \int_{\mathcal{M}} d^4x \sqrt{|g|} \tilde{R}_{\lambda \rho}^{\mu \nu} R_{\mu \nu}^{\lambda \rho}. \quad (115)$$

This invariant represents the gravitational analog of Eqs. (26) and (50) for $U(1)$ and $SU(2)$, respectively. Although this term can be written as a boundary term introducing no modifications to the bulk dynamics of the metric, it yields finite contribution to the action and variations thereof in asymptotically locally AdS spacetimes. Indeed, it can be added to the action with a fixed coupling constant such that the ground state of the theory is enlarged to all the (anti-)self-dual configurations [7, 8]. We will study this in the following section.

C. (Anti-)Self duality of the displaced Riemann tensor

In Riemannian geometry, it is possible to define (anti-)self-duality at the level of the Riemann curvature tensor or, in the presence of a (negative) cosmological constant, in terms of the displaced Riemann tensor in Eq. (102), that is,

$$F_{\mu \nu \lambda \rho} = \pm \tilde{F}_{\mu \nu \lambda \rho}, \quad \text{where} \quad \tilde{F}_{\mu \nu \lambda \rho} = \frac{1}{2} \varepsilon_{\mu \nu \alpha \beta} F_{\lambda \rho}^{\alpha \beta}, \quad (116)$$

is the dual F -tensor. Configurations satisfying this condition bear close resemblance with instantons in Yang-Mills theory. For instance, if Eq. (116) is satisfied, then Einstein's field equations are automatically by virtue of Bianchi's identities, as

$$\begin{aligned}\mathcal{E}_\nu^\mu &= -\frac{1}{4}\delta_{\nu\lambda\rho}^{\mu\alpha\beta}F_{\alpha\beta}^{\lambda\rho} = \mp\frac{1}{4}\delta_{\nu\lambda\rho}^{\mu\alpha\beta}\tilde{F}_{\alpha\beta}^{\lambda\rho} = \mp\frac{1}{8}\left(\delta_\nu^\mu\delta_{\lambda\rho}^{\alpha\beta} + \delta_\rho^\mu\delta_{\nu\lambda}^{\alpha\beta} + \delta_\lambda^\mu\delta_{\rho\nu}^{\alpha\beta}\right)\varepsilon^{\lambda\rho\sigma\tau}F_{\sigma\tau\alpha\beta} \\ &= \mp\frac{1}{4}\left(\delta_\nu^\mu\varepsilon^{\lambda\rho\sigma\tau}F_{\sigma\tau\lambda\rho} + \varepsilon^{\lambda\mu\sigma\tau}F_{\sigma\tau\nu\lambda} + \varepsilon^{\mu\rho\sigma\tau}F_{\sigma\tau\rho\nu}\right) \\ &= 0,\end{aligned}\tag{117}$$

since $F_{\mu[\nu\lambda\rho]} = 0$. Thus, Eq. (116) implies that (anti-)self-dual displaced Riemann configurations are Einstein spaces. This, in turn, implies that the Weyl tensor is (anti-)self-dual by virtue of Eq. (107), that is, $W_{\mu\nu\lambda\rho} = \pm\tilde{W}_{\mu\nu\lambda\rho}$. Indeed, using the definition of the displaced Riemann tensor in Eq. (102) and replacing the (anti-)self-dual condition (116) into the topologically renormalized action (113), one obtains

$$\begin{aligned}I_{\text{EH}}^{(\text{ren})} &= \frac{\kappa\ell^2}{16}\int_{\mathcal{M}}d^4x\sqrt{|g|}\delta_{\nu_1\dots\nu_4}^{\mu_1\dots\mu_2}F_{\mu_1\mu_2}^{\nu_1\nu_2}F_{\mu_3\mu_4}^{\nu_3\nu_4} \\ &= \pm\frac{\kappa\ell^2}{4}\int_{\mathcal{M}}d^4x\sqrt{|g|}F_{\lambda\rho}^{\mu\nu}\tilde{F}_{\mu\nu}^{\lambda\rho} \\ &= \pm\frac{\kappa\ell^2}{4}\int_{\mathcal{M}}d^4x\sqrt{|g|}R_{\lambda\rho}^{\mu\nu}\tilde{R}_{\mu\nu}^{\lambda\rho} \\ &= \pm 4\pi^2\kappa\ell^2 P_1[\mathcal{M}],\end{aligned}\tag{118}$$

where, in the second line, we have used the identity

$$\delta_{\nu_1\dots\nu_4}^{\mu_1\dots\mu_4} = \delta_{\nu_1\nu_2}^{\mu_1\mu_2}\delta_{\nu_3\nu_4}^{\mu_3\mu_4} + \delta_{\nu_1\nu_4}^{\mu_1\mu_2}\delta_{\nu_2\nu_3}^{\mu_3\mu_4} + \delta_{\nu_1\nu_3}^{\mu_1\mu_2}\delta_{\nu_4\nu_2}^{\mu_3\mu_4} + \delta_{\nu_2\nu_3}^{\mu_1\mu_2}\delta_{\nu_1\nu_4}^{\mu_3\mu_4} + \delta_{\nu_2\nu_4}^{\mu_1\mu_2}\delta_{\nu_3\nu_1}^{\mu_3\mu_4} + \delta_{\nu_3\nu_4}^{\mu_1\mu_2}\delta_{\nu_1\nu_2}^{\mu_3\mu_4},\tag{119}$$

and, in the third line, we have used $F_{\lambda\rho}^{\mu\nu}\tilde{F}_{\mu\nu}^{\lambda\rho} = R_{\lambda\rho}^{\mu\nu}\tilde{R}_{\mu\nu}^{\lambda\rho}$. Therefore, upon evaluation on (anti-)self-dual configurations, the renormalized action becomes a topological invariant of the Pontryagin class; this is analogous to what happens with instantons in Yang-Mills theories. Indeed, one can modify the action by adding the Chern-Pontryagin index of $SO(4)$ without changing the bulk dynamics. However, if the coupling of this topological term is chosen such that

$$\bar{I}_{\text{EH}}^{(\text{ren})} = \frac{\kappa\ell^2}{16}\int_{\mathcal{M}}d^4x\sqrt{|g|}\left[\delta_{\nu_1\dots\nu_4}^{\mu_1\dots\mu_4}\left(R_{\mu_1\mu_2}^{\nu_1\nu_2} + \frac{1}{\ell^2}\delta_{\mu_1\mu_2}^{\nu_1\nu_2}\right)\left(R_{\mu_3\mu_4}^{\nu_3\nu_4} + \frac{1}{\ell^2}\delta_{\mu_3\mu_4}^{\nu_3\nu_4}\right) \mp 4R_{\lambda\rho}^{\mu\nu}\tilde{R}_{\mu\nu}^{\lambda\rho}\right],\tag{120}$$

the action vanishes identically on (anti-)self-dual configurations in the sense of Eq. (116). Thus, the ground state of the theory is enlarged to all this family of solutions. This set, of course,

contains maximally symmetric spaces, but there are non-constant curvature spaces satisfying the (anti-)self-dual condition. These are known as gravitational instantons, and we will present the main features of a particular one, the Euclidean Taub-NUT-AdS metric, in the following section.

D. Euclidean (anti-)self-dual Taub-NUT-AdS instanton

The Euclidean (anti-)self-dual Taub-NUT-AdS solution can be obtained by considering a $U(1)$ fibration over Einstein-Kähler manifolds. In particular, if the latter is taken to be $\mathbb{S}^2$, its line element can be parametrized in terms of Euler angles as

$$ds^2 = f(r)(d\tau + 2n \cos \vartheta d\varphi)^2 + \frac{dr^2}{f(r)} + (r^2 - n^2) (d\vartheta^2 + \sin^2 \vartheta d\varphi^2), \quad (121)$$

where n is dubbed the NUT charge. Indeed, after rescaling the Euclidean time as $\tau \rightarrow \tau/(2n)$, it can be written in terms of the left-invariant forms of $SU(2)$ in Eq. (44) as

$$ds^2 = 4n^2 f(r) \sigma_3^2 + \frac{dr^2}{f(r)} + (r^2 - n^2) (\sigma_1^2 + \sigma_2^2) = \frac{dr^2}{f(r)} + (r^2 - n^2) \left(\sigma_1^2 + \sigma_2^2 + \frac{4n^2 f(r)}{r^2 - n^2} \sigma_3^2 \right).$$

Thus, one concludes that the hypersurfaces of constant r represent squashed spheres, whose squashing parameter depends on the value of the NUT charge and $f(r)$ at a given radius.

We look for configurations satisfying (anti-)self-duality at the level of the displaced Riemann tensor since they satisfy the Einstein field equations (104) automatically. Using the ansatz (121) and solving Eq. (116), we find that the metric function is given by

$$f(r) = \frac{r - n}{r + n} + \frac{(r - n)^2 (r + 3n)}{\ell^2 (r + n)}. \quad (122)$$

This metric does not have curvature singularities for $n \leq r < \infty$. However, like the magnetic monopole, the Taub-NUT metric has a string-like singularity at $\vartheta = 0, \pi$. This can be seen by choosing a vierbein basis

$$e^0 = \sqrt{f(r)} (d\tau + 2n \cos \vartheta), \quad e^1 = \frac{dr}{\sqrt{f(r)}}, \quad e^2 = \sqrt{r^2 - n^2} d\vartheta, \quad e^3 = \sqrt{r^2 - n^2} \sin \vartheta d\varphi, \quad (123)$$

and noticing that the norm of the vector field that generates the Euclidean time evolution is not well-defined at $f(r) = 0$ and $\vartheta = 0, \pi$, since

$$\nabla_\mu \tau \nabla^\mu \tau = \frac{1}{f(r)} + \frac{4n^2 \cot^2 \vartheta}{r^2 - n^2}. \quad (124)$$

This is usually called the Misner string, which represents the gravitational analog of the Dirac string. Indeed, in Ref. [9], it was shown that its position can be rendered unobservable by performing an improper coordinate transformation, i.e. $\tau = \bar{\tau} + 2nC\varphi$ with $C = \pm 1, 0$ being a constant, giving

$$ds^2 = f(r) (d\bar{\tau} + 2n(\cos \vartheta + C)d\varphi)^2 + \frac{dr^2}{f(r)} + (r^2 - n^2) (d\vartheta^2 + \sin^2 \vartheta d\varphi^2) , \quad (125)$$

with $f(r)$ given in Eq. (122). If $C = \pm 1$, then the Misner string is located at $\vartheta = 0$ or $\vartheta = \pi$, respectively; the case $C = 0$ leads to the initial line element (121).

In Lorentzian signature, however, this solution is endowed with closed time-like curves [9]. This can be seen by noticing that, since φ is periodic, $\bar{\tau}$ will be identified at the equator. Thus, rendering the position of the Misner string unobservable leads to closed time-like curves in the Lorentzian section. In Euclidean signature, there is no such a problem as all curves are spacelike. Nevertheless, one must eliminate conical singularities at the zero-dimensional set of fixed points. This can be done by demanding $\tau \sim \tau + \beta$, where the period of the Euclidean time is

$$\beta = \left. \frac{4\pi}{f'(r)} \right|_{r=n} = 8\pi n . \quad (126)$$

Then, if $n < r < \infty$, $0 \leq \tau \leq 8\pi n$, $0 \leq \vartheta \leq \pi$, and $0 \leq \varphi < 2\pi$, the line element (121) with metric function (122) is a completely regular space with (anti-)self-dual Weyl tensor, whose action is proportional to the Chern-Pontryagin index of $SO(4)$. In this case, we find that

$$I_{\text{EH}}^{(\text{ren})} = \pm 4\pi^2 \kappa \ell^2 P_1[\mathcal{M}] = \pm 8\pi^2 \kappa \ell^2 \left(1 - \frac{4n^2}{\ell^2} \right) , \quad (127)$$

which saturates the BPS bound. Additionally, the Euler characteristic can be computed directly from the Gauss-Bonnet theorem in Eq. (108), giving $\chi(\mathcal{M}) = 1$. The index of the Dirac operator was computed in Ref. [54], showing that the (anti-)self dual Taub-NUT instanton does not contribute to the axial anomaly of Dirac fermions, even though it does contribute when Rarita-Schwinger spinors are considered [54]. The conserved charges of this solution using the Noether-Wald formalism, alongside topological renormalization, have been studied in Refs. [8, 10, 55]

E. Holographic stress-energy tensor

The gauge/gravity duality has proven to be a remarkable tool for studying strongly coupled quantum systems through classical gravitational theories in asymptotically AdS spacetimes [56–58].

For instance, the vacuum expectation value of the holographic stress-energy tensor can be obtained from the renormalized variation of a gravitational action with suitable boundary conditions. To do this, counterterms must be added to the action such that its value and variations thereof become finite on asymptotically locally AdS spacetimes. Then, one can read off the holographic stress-energy tensor of the dual theory from a rescaled version of the renormalized quasilocal stress tensor of the boundary metric in AdS spacetimes. Let us compute the holographic stress-energy tensor for the (anti-)self dual Taub-NUT-AdS instanton.

In asymptotically locally AdS spacetimes, the metric can be expanded towards the asymptotic boundary in a Fefferman-Graham form, that is,

$$ds^2 = \frac{\ell^2}{z^2} (dz^2 + \bar{g}_{ij}(z, x) dx^i dx^j) , \quad (128)$$

where the boundary is located at $z = 0$. In general, $\bar{g}_{ij}$ admits a series expansion in the holographic coordinate as

$$\bar{g}_{ij}(z, x) = g_{(0)ij}(x) + \frac{z}{\ell} g_{(1)ij}(x) + \frac{z^2}{\ell^2} g_{(2)ij}(x) + \frac{z^3}{\ell^3} g_{(3)ij}(x) + \mathcal{O}(z^4) , \quad (129)$$

where $g_{(0)ij}$ is the boundary metric. For Einstein spaces, however, only the odd powers in z survive; this results from solving the field equations order by order in the holographic coordinate.

Performing arbitrary variations of the renormalized action (113), expanding the result using the Fefferman-Graham gauge (128), and using the coefficients of $\bar{g}_{ij}$ obtained by solving the field equations, one obtains (see Ref. [7] for details)

$$\delta I_{\text{EH}}^{(\text{ren})} = \int_{\partial\mathcal{M}} d^3x \sqrt{|g_{(0)}|} \delta g_{(0)}^{ij} \tau_{ij} , \quad \text{where} \quad \tau_{ij} = -\frac{\ell}{8\pi G} \mathcal{E}_{ij} \quad (130)$$

is the holographic stress-energy tensor, defined in terms of the electric part of the Weyl tensor $\mathcal{E}_{ij}$. The latter, alongside its magnetic part, are given by

$$\mathcal{E}_{ij} = W_{i\lambda j\rho} n^\lambda n^\rho \quad \text{and} \quad \mathcal{B}_{ij} = \tilde{W}_{i\lambda j\rho} n^\lambda n^\rho , \quad (131)$$

respectively. To relevant holographic order, the magnetic part of the Weyl tensor can be expressed in terms of the boundary Cotton tensor as [7, 59, 60]

$$-\ell C_{ij}(g_{(0)}) = \mathcal{B}_{ij} , \quad \text{where} \quad C^{ij}(g_{(0)}) = \varepsilon^{(0)ilm} \nabla_{(0)m} (R_{(0)l}^j - \frac{1}{4} R_{(0)} \delta_l^j) \quad (132)$$

is the Cotton tensor of the boundary metric; all quantities in the last expression are constructed out of $g_{(0)}$.

In asymptotically locally AdS spaces, the (anti-)self-duality condition in Eq. (116) is translated into $\mathcal{E}_{ij} = \pm \mathcal{B}_{ij}$. Then, the holographic stress-energy tensor for the (anti-)self-dual Taub-NUT-AdS instanton in Eq. (122) can be written as [10]

$$\tau_{ij} = \pm \frac{\ell^2}{8\pi G} C_{ij} \quad \text{where} \quad C_{ij} = \frac{n}{\ell^3} \left(1 - \frac{4n^2}{\ell^2} \right) \sigma_{ij}(\vartheta), \quad (133)$$

is the boundary Cotton tensor, and we have defined the codimension-1 boundary tensor as

$$\sigma_{ij}(\vartheta) = \begin{pmatrix} 2 & 0 & 4n \cos \vartheta \\ 0 & \ell^2 & 0 \\ 4n \cos \vartheta & 0 & \ell^2 \sin^2 \vartheta + 8n^2 \cos^2 \vartheta \end{pmatrix}, \quad (134)$$

with the orientation $(\tau, \vartheta, \varphi)$. Similar to the electric/magnetic part of the Weyl tensor, the boundary Cotton tensor is traceless with respect to the boundary metric. In the dual theory, this implies that the trace of the vacuum expectation value of the holographic stress tensor vanishes, producing no conformal anomaly in three dimensions. Moreover, the Cotton tensor is proportional to the NUT charge. The latter is a squashing parameter that deforms the boundary, rendering it non-conformally flat. Indeed, this form of the holographic stress tensor is useful for describing fluids with nontrivial vorticity, as shown in Refs. [61–63]. We encourage the interested reader to review those works.

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